

Poisson Regression

Data

2015 Family Income and Expenditure Survey (FIES) on households in the Philippines. Variables include

- + `age`: age of the head of household
- + `numLT5`: number in the household under 5 years old
- + `total`: total number of people other than head of household
- + `roof`: type of roof (stronger material can sometimes be used as a proxy for greater wealth)
- + `location`: where the house is located (Central Luzon, Davao Region, Ilocos Region, Metro Manila, or Visayas)

Data

Questions:

- + How is the age of head of household related to the number of people in the household?
- + Is the type of roof material related to the number of people in the household?

To answer these questions, our response variable is total (total number of people other than head). What kind of variable is this?

- quantitative, discrete, values 0, 1, 2, 3, ...

=> A count variable

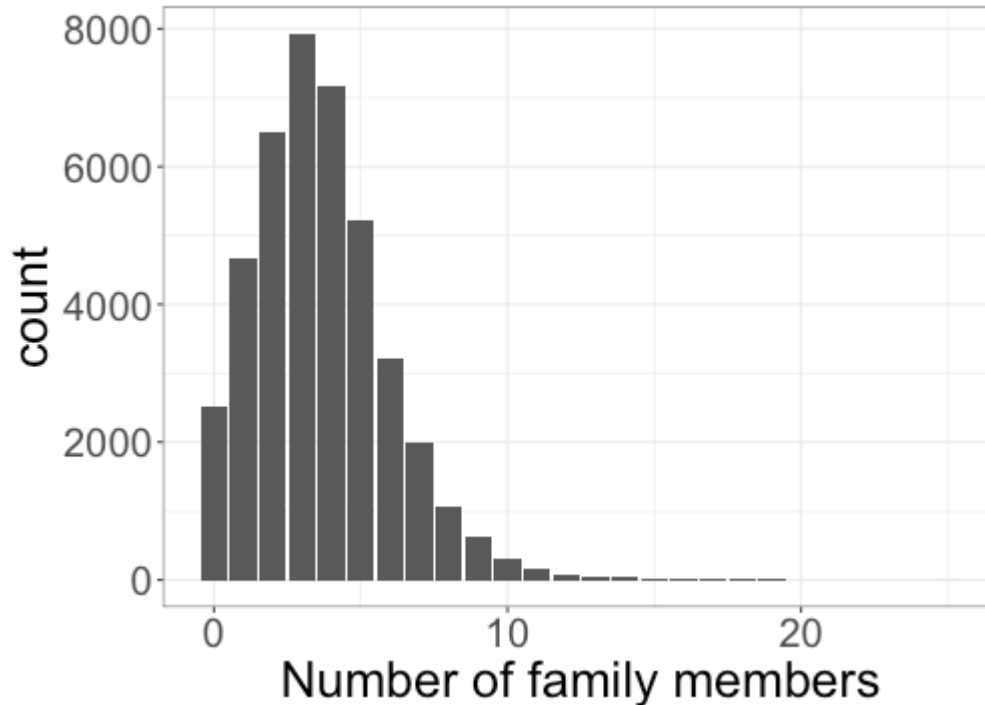
Building a parametric model

Step 1: Choose a reasonable distribution for Y

- + $Y_i = total_i$ is a count variable!
- + Unfortunately, we don't know any distributions for count data
- + Bernoulli and Categorical distributions are for categorical data
- + Normal distributions are for continuous data; count data is discrete

We need a new distribution!

Exploring the response



- + Right skewed, unimodal distribution
- + We can use a *Poisson* distribution

Poisson distribution

If $Y_i \sim \text{Poisson}(\lambda)$, then Y_i takes values $y = 0, 1, 2, \dots$ with probabilities

$$P(Y_i = y) = \frac{e^{-\lambda} \lambda^y}{y!}$$

Does this distribution look familiar?

Poisson distribution

$\sim \text{Poisson}(\lambda)$

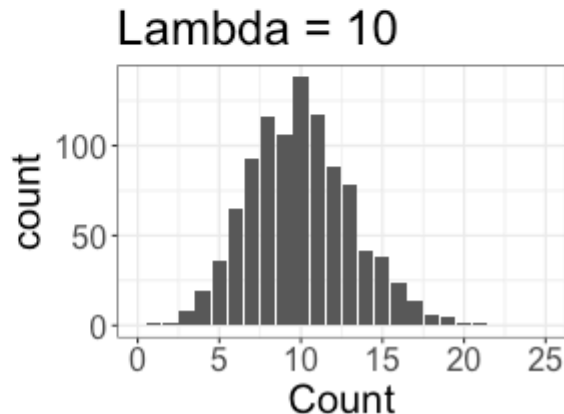
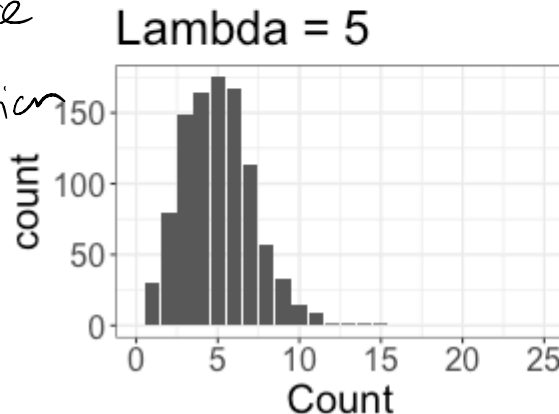
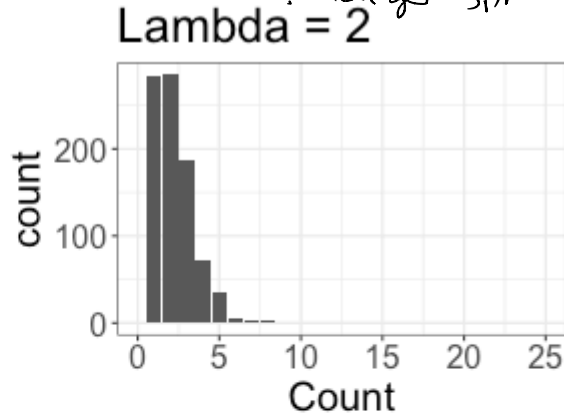
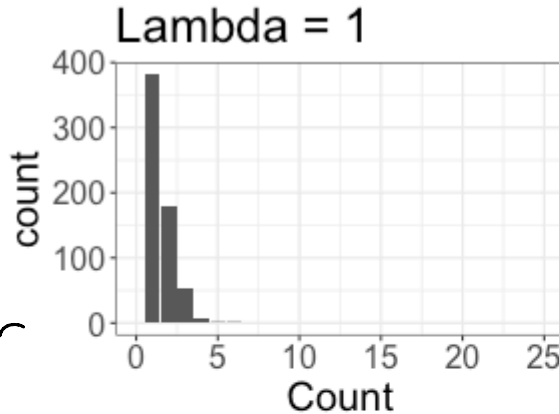
Larger $\lambda \Rightarrow$

- less skewed
- larger values of λ (higher variability)
- larger spread

centered
at λ

$\lambda = \text{mean of distribution}$

$\lambda = \text{variance of distribution}$



How is λ related to the distribution?

Poisson distribution

If $Y_i \sim \text{Poisson}(\lambda)$, then Y_i takes values $y = 0, 1, 2, \dots$ with probabilities

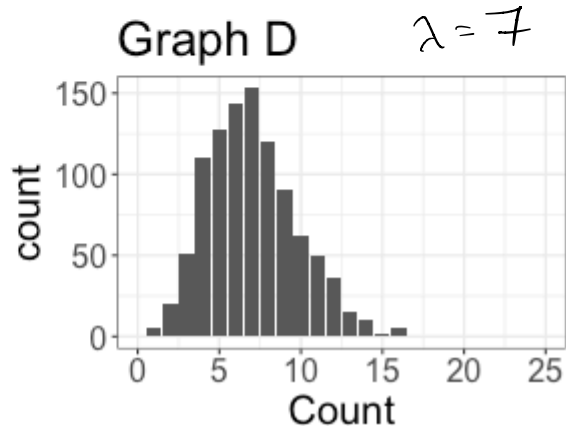
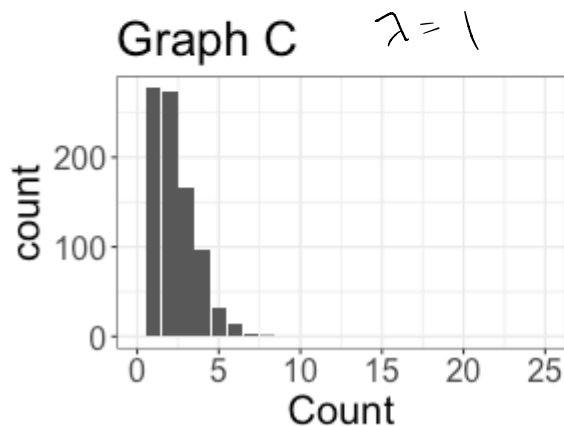
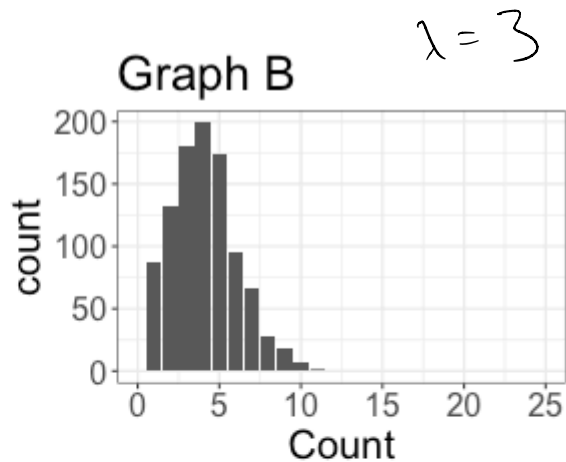
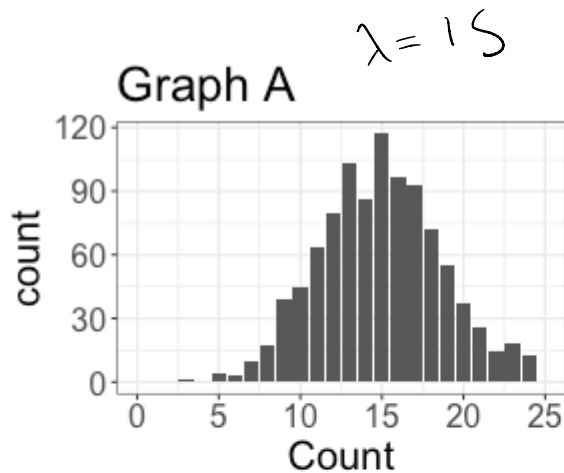
$$P(Y_i = y) = \frac{e^{-\lambda} \lambda^y}{y!}$$

- + λ is the mean of the distribution
- + λ is also the variance! (the mean and variance are the same)
- + Our goal is to estimate λ , just like our goal was to estimate π in logistic regression

Class activity

Work on the activity (handout), then we will discuss as a class.

Class activity



What do you think λ is for each graph?

Class activity

Y_i = number of dogs adopted from animal shelter

$$Y_i \sim \text{Poisson}(1.5)$$

What is the probability that at most two dogs are adopted?

$$\begin{aligned} & P(0) + P(1) + P(2) \\ &= \frac{e^{-1.5} (1.5)^0}{0!} + \frac{e^{-1.5} (1.5)^1}{1!} + \frac{e^{-1.5} (1.5)^2}{2!} \\ &= 0.81 \end{aligned}$$

Class activity

The Poisson distribution is for count data. Why is it ok for λ to not be a whole number?

λ represents the mean count, which
can be a fraction

Poisson regression

Y_i = number of people in household other than head

How is Y_i related to the age of the head of the household?

Step 1: Choose a reasonable distribution for Y

$$Y_i \sim \text{Poisson}(\lambda_i)$$

← random component

Step 2: Choose a model for any parameters

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Age}_i$$

← systematic component

Why do you think we use $\log(\lambda_i)$ instead of just λ_i ?

$$\lambda_i > 0, \quad \text{but } \log(\lambda_i) \in (-\infty, \infty)$$
$$\beta_0 + \beta_1 x_i \in (-\infty, \infty)$$

Fitting the model

distribution of
response

```
m1 <- glm(total ~ age, family = poisson,  
           data = fies)  
summary(m1)
```

...

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

```
##
```

```
## (Dispersion parameter for poisson family taken to be 1)
```

...

$$\log(\hat{\lambda}_i) = 1.714 - 0.0035 \text{ Age}_i$$

How can we interpret the slope?

Fitting the model

$$\log(\hat{\lambda}_i) = 1.714 - 0.0035 \text{ Age}_i$$

For every additional year in age of the head of house, we expect the log of the *average* household size to decrease by 0.0035.

Can I interpret on the un-logged scale?

$$e^{-0.0035} = 0.9965$$

For each additional year in age, we expect average household size changes by a factor of 0.9965 (a decrease of 0.35%)

Assumptions

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Age}_i$$

What assumptions does our Poisson regression model make?

- Distribution: Y_i really does follow a Poisson distribution
- Independence $\leftarrow$ need for MLE
- Randomness (data come from same random process)
- Shape: the shape of the regression model is correct

Assumptions

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Age}_i$$

What assumptions does our Poisson regression model make?

- + **Shape:** The shape of the regression model is correct
- + **Independence:** The observations are independent
- + **Poisson distribution:** A Poisson distribution is a good choice for Y_i

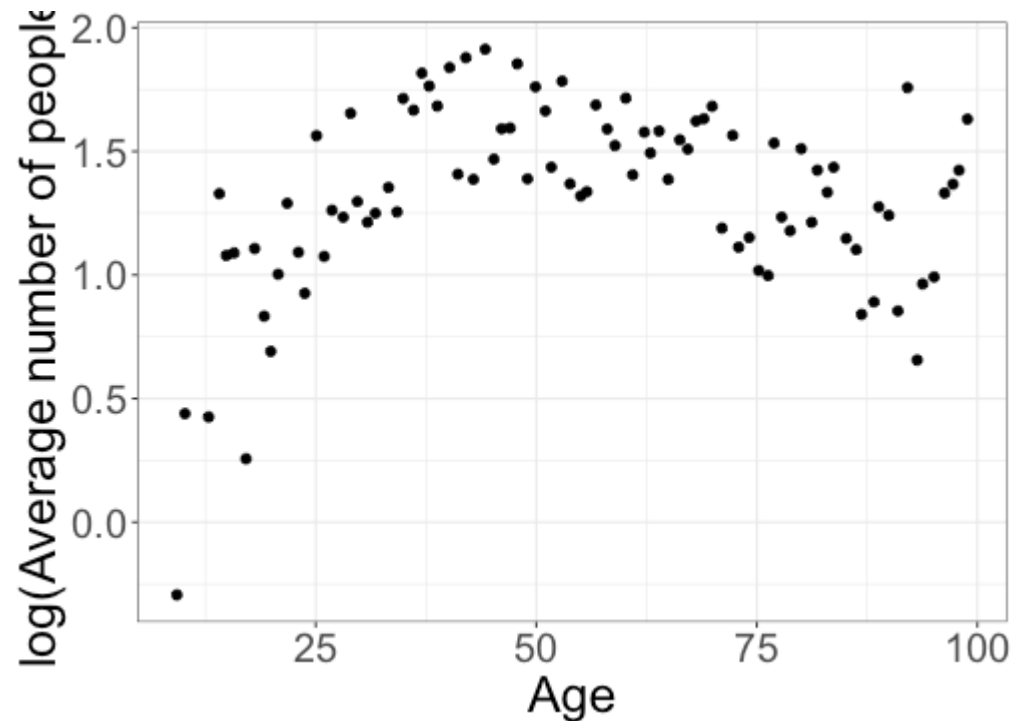
The shape assumption

Shape assumption: The shape of the regression model is correct

How can I assess this assumption?

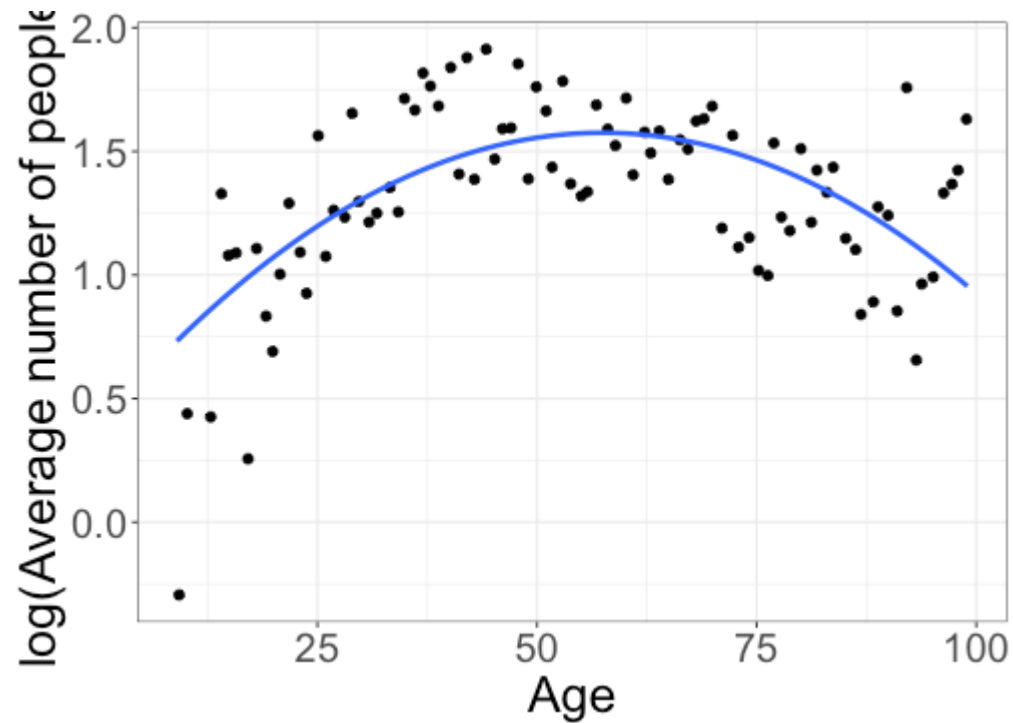
- Quantile residual plots
- empirical log mean plot
 - ↳ Take an explanatory variable
 - Divide it into bins
 - calculate $\log(\bar{Y})$ in each bin
 - Plot

Checking the shape assumption



What shape seems appropriate?

Second order polynomial



$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Age}_i^2$$

Poisson distribution assumption

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{Age}_i + \beta_2 \text{Age}_i^2$$

Poisson distribution assumption: The Poisson distribution is a good choice for Y_i .

What are some characteristics of the Poisson distribution we could check?

Checking distribution shape

Look at the distribution of Y_i for different ages:

